

$$\begin{aligned}x + ky &= 1 \\ kx + y &= -1\end{aligned}$$

Augmented Matrix:

$$\begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & k & 1 \\ k & 1 & -1 \end{array} \right]$$

$$R_2 \rightarrow R_2 + R_1$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & k & 1 \\ k+1 & k+1 & 0 \end{array} \right] = [A|B]$$

For Unique $\Rightarrow |A| \neq 0 \Rightarrow \begin{vmatrix} 1 & k \\ k & 1 \end{vmatrix} \neq 0$

$$\Rightarrow k \neq 1, -1$$

For infinite $\Rightarrow$

$$\left[\begin{array}{cc|c} 1 & k & 1 \\ k+1 & k+1 & 0 \end{array} \right] \Rightarrow k+1 = 0 \Rightarrow k = -1$$

For no solutⁿ $\Rightarrow |A| = 0$ and any ~~rank~~ Rank $([A|B]) = 2$

$$\Rightarrow k = 1$$